



AI Tutor Agent

User Guide & Manual

Version 2.1 · 2026

ai-tutor-agent-ten.vercel.app

RAG + Streaming

5 Learning Modes

3 AI Providers

Introduction

AI Tutor Agent is a cloud-based learning assistant powered by Retrieval-Augmented Generation (RAG). Upload any document — PDF, Word, Markdown, image, audio, or video — and the app builds a searchable knowledge base. Ask questions in plain English and receive streamed, source-cited answers from Claude, Gemini, or OpenAI.

■ Live App

Access the app at: ai-tutor-agent-ten.vercel.app — no installation required.

Quick Start

1	Open the App Go to ai-tutor-agent-ten.vercel.app in any modern browser.
2	Upload a Document Click the ■ File button in the sidebar or drag & drop a file onto the upload zone.
3	Wait for Processing A progress bar shows upload %, then the app processes the content into the knowledge base.
4	Choose a Mode Select Explain, Quiz, Chat, Summarize, or Flashcards from the top toolbar.
5	Ask a Question Type your question in the chat box and press Send (or Enter).
6	Get Your Answer The AI streams a markdown-formatted reply citing which document answered it.

Uploading to the Knowledge Base

The sidebar contains the upload panel. Two methods are available:

■ File Upload

PDF	Full text extraction from all pages
Word (.docx)	Raw text extracted (formatting stripped)
Markdown / TXT	Ingested as-is
Images (JPG/PNG/GIF/WEBP)	Described in detail by Claude or GPT-4o vision
Audio (MP3/WAV/M4A)	Transcribed with OpenAI Whisper
Video (MP4/MOV/AVI)	Audio track transcribed with Whisper

■ URL Ingest

Switch to the URL tab and paste any public web page or PDF link. The app fetches and processes the content automatically.

■ Progress Bar

A live % bar tracks the upload. Once 100% is reached, the bar pulses while the AI processes (vision / transcription). Large video files may take up to 60 s.

The 5 Learning Modes

■ Explain	■ Quiz	■ Chat	■ Summarize	■ Flashcards
In-depth breakdown with examples and analogies. Ends with a comprehension check.	Generates one focused question. Provides feedback after you answer.	Free-form conversation. Great for follow-up questions.	Structured overview with headings, bullets, and Key Takeaways.	Returns 5 Q&A; pairs formatted for quick revision.

Switch modes at any time using the toolbar above the chat. Each mode uses a tailored system prompt optimised for that learning style.

AI Providers

■ Claude (Agent 1)	Default. Best for creative explanation, tutoring, and nuanced writing.
■ Gemini (Agent 2)	Best for broad general knowledge, multi-topic questions.
■ OpenAI (Agent 3)	Best for analytical reasoning, structured problem-solving.

Switch providers using the AI Provider panel in the sidebar or by voice: "Hey Agent 1", "Hey Agent 2", or "Hey Agent 3". Auto-fallback is enabled — if one provider is unavailable, the app automatically tries the next.

■ Auto-Switch on Role Change

When you select a Role, the app automatically switches to the best-fit provider (e.g. Brainy Expert → OpenAI; General Knowledge → Gemini; others → Claude).

Roles & Personas

Use the Role dropdown (top-right of the chat panel) to change the AI's persona. Each role adapts tone, depth, and style:

■ AI Tutor	Socratic teaching style. Asks guiding questions. Scaffolds explanations.
■ Receptionist	Friendly, concise. Direct answers. 2–4 sentences. No over-explaining.
■ Brainy Expert	Rigorous analysis. Cites evidence. Distinguishes fact from inference.
■ General Knowledge Agent	Encyclopaedic breadth. Neutral language. Connects facts across domains.
■ Creative Assistant	Imaginative. Varied sentence rhythm. Generates 3+ directions then refines.
■■ Custom role	Type any persona up to 80 characters (e.g. "Pirate Captain", "Finance CFO").

Features Reference

■ Chat Persistence All messages are saved to localStorage keyed by session UUID. Survive page refresh automatically.	■ Mobile Sidebar Tap ■ in the header to open the sidebar as a full-height overlay. Tap outside to close.	■ Code Copy Button Hover over any code block to reveal a Copy button in the top-right corner.
■■ KB Reset Warning Amber banner appears after server restart wipes the in-memory vector store.	■■ Voice Dictation Click ■■ Dictate to speak your question. Say "Hey Agent 1/2/3" to switch providers.	■ Auto-Read Toggle Auto-read ON/OFF. Click Read answer to replay the last response via TTS.
■■ Image Attachment Attach an image to your message for visual analysis alongside your question.	■ Export Chat Download the full conversation as a Markdown file via the ↓ button in the toolbar.	■■ Stop Generation Click the stop button during streaming to abort the response mid-reply.

Managing the Knowledge Base

View docs	The Knowledge Base panel (sidebar) shows chunks, doc count, and a list of sources.
Delete a doc	Click ■■ next to any document name to remove it and its chunks immediately.
Re-upload	After a server restart (cold start), re-upload documents — they are stored in RAM.
Clear history	Click the ■ button in the chat toolbar to clear chat and start a new session.

Keyboard Shortcuts

Enter	Send message
Shift + Enter	New line in message (no send)
Escape	Cancel current voice dictation

Tips & Best Practices

■ Use Explain for new topics

Start every new subject in ■ Explain mode. The AI scaffolds from simple to complex and ends with a comprehension check.

■ Quiz yourself after reading

Switch to ■ Quiz after an Explain session. The AI generates one targeted question per message — answer it, then press Send for the next.

■ Flashcards for revision

Use ■ Flashcards when reviewing. Copy the 5 Q&A; pairs and paste into Anki or Notion for spaced repetition.

■ Focus on a specific doc

If multiple documents are loaded, mention the filename in your question — the app prioritises that source first.

■ Richer answers with images

Attach a diagram or screenshot to your message. Claude and GPT-4o will describe it and incorporate it into the answer.

■ Ingest audio lectures

Upload MP3/WAV recordings. Whisper transcribes them into the knowledge base — then quiz yourself on the lecture content.

■ Custom role for specialised help

Set a custom role like "Tax Accountant UK" or "Python Senior Engineer" for domain-specific answers.

■ Keep the KB fresh

Vercel serverless functions restart on cold start, wiping the in-memory vector store. Re-upload documents if the KB reset warning appears.

Troubleshooting

■■ 'Knowledge base was reset'	Vercel restarted the server. Re-upload your documents — the vector store is RAM-only.
■■ '502 AI provider error'	API key is missing or invalid. Check environment variables in Vercel Settings.
■■ Backend offline message	The serverless function is cold-starting. Refresh in 5–10 seconds.
No audio from TTS	Browser may block autoplay. Click the ■ Read answer button manually.
Dictation not working	Grant microphone permission when the browser prompts. Use HTTPS (required for mic API).
Large video fails to upload	Videos over 50 MB may time out on Vercel. Compress first or extract the audio track.
Chat history lost	Chat is saved per session UUID. Clearing browser localStorage will erase history.

API Key Configuration

For self-hosted deployments, add these environment variables in Vercel → Project → Settings → Environment Variables:

OPENAI_API_KEY	Required always — powers text-embedding-3-small for RAG retrieval.
LLM_PROVIDER	Set to claude, openai, or gemini (default: claude).
CLAUDE_API_KEY	Required when LLM_PROVIDER=claude (Anthropic key).
GEMINI_API_KEY	Required when LLM_PROVIDER=gemini (Google AI key).
ALLOWED_ORIGIN	Set to your Vercel app URL for CORS (e.g. <code>https://your-app.vercel.app</code>).

■ Support

For issues, feature requests, or contributions, visit the GitHub repository: github.com/paulcoleetd/AT-Tutor-2